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ALGAL ZONATION IN THE NEW ENGLAND ROCKY INTERTIDAL COMMUNITY: AN EXPERIMENTAL ANALYSIS[1]

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Abstract. Zonation patterns of plants, including marine algae, have commonly been attributed solely and directly to physical factors. Experimental investigations of the factors affecting zonation of macroscopic, benthic algae in the New England rocky intertidal region demonstrate that biological factors set the lower limits of these plants. The mid zone at all but very exposed sites is usually dominated by brown fucoid algae. These plants are virtually absent in the low zone, which is dominated by the red alga *Chondrus crispus* (Irish moss). Total removal of *Chondrus* (including the prostrate holdfast) results in establishment of *Fucus vesiculosus* or *F. distichus* ssp. *edendatus* in the low zone. *Fucus* grows faster in the low than in the mid zone, appears healthy, and reproduces. Thus competition from *Chondrus* sets the lower limit of *Fucus*, not changes in light intensity or immersion time, per se, as previously assumed. If herbivores (primarily the periwinkle snail *Littorina littorea*) are absent where *Chondrus* is removed, *Fucus* can settle very densely and occupy 100% of the space. If herbivores are present, *Fucus* colonizes, but less abundantly. Thus competition is the primary determinant of the zonation pattern (affecting presence or absence) and herbivory is of secondary importance (affecting abundance).

Other experimental evidence suggests that the upper limit of *Chondrus* is determined by desiccation. The lower limit of *Chondrus* has not been investigated except where a sharp lower limit exists at the low intertidal-shallow subtidal interface. Experiments demonstrate that this is due to the grazing by sea urchins (*Strongylocentrotus droebachiensis*) where they are locally abundant. Normally, *Chondrus* extends well into the subtidal region.

These results parallel experimental studies of animal zonation in rocky intertidal regions in which biotic factors also set lower bathymetric limits. It is suggested that biogeographic ranges of some species may be similarly affected by biotic factors.

Key words: algae; Chondrus; *community structure; competition;* Fucus; *herbivory; New England; niche width; zonation.*

INTRODUCTION

Zonation of plants and animals along environmental gradients is a universal phenomenon. Particularly striking are zonation patterns along rocky intertidal shores. Until the last two decades, studies of rocky intertidal plant and animal zonation were largely descriptive (e.g., Stephenson and Stephenson 1972). The correlations of the zones with changes in emergence time, light quality or intensity were assumed to be causal (Gail 1918, Coleman 1933, Hewatt 1937, Doty 1946, Southward 1958, Lewis 1964, Newell 1970; but see Underwood 1978). Beginning with Connell's (1961*a*, *b*) studies of factors affecting zonation of barnacles, experimental investigation of rocky intertidal animals has demonstrated unequivocally that lower limits of these organisms are usually determined by biological interactions, either predation or competition (Connell 1972). Despite the parallels between sessile animals and plants, and suggestions by numerous workers that competition and herbivory probably affect lower limits of algae (Chapman 1957, Southward 1958, Kitching and Ebling 1961, Lewis 1964, Jones and Kain 1967, Vadas 1968, Connell 1972, Chapman 1973, 1974, Hruby 1976, and Schonbeck and Norton 1978), zonation of marine plants is still often assumed to be solely a function of physical factors: light intensity, light quality, emergence time, submergence time, etc.

Studies of factors affecting zonation of algae may be interesting for a number of reasons. First, there are strong parallels between studies of bathymetric zonation in marine plants and studies of altitudinal, soil moisture, and soil type zonation in terrestrial vegetation. Both have remained largely descriptive (with a few notable exceptions, e.g., Vadas 1968, Dayton 1975, Schonbeck and Norton 1978, and Hruby 1976 for marine algae and Sharitz and McCormick 1973 for terrestrial plants). The majority of terrestrial zonation investigations assume that some abiotic factor that changes along the gradient causes the species replacement patterns because species are specialized to distinct parts of the gradient. Studies are then initiated to isolate the factor or group of factors that best correlates with the vegetation changes. For example, the striking zonation of tree species along an altitudinal gradient in the Green Mountains of Vermont is suggested to be caused by a climatic discontinuity resulting in a sudden change in the number of frost-free days and the amount of moisture and hoar frost present (Siccama 1974). This conclusion is

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remarkably similar to the suggestion that zonation of marine algae is caused by discontinuities in submergence times, or so-called "critical tide factors" due to the pecularities of tidal cycles (Coleman 1933, Doty 1946). Given the similarities between these marine and terrestrial patterns perhaps experimental investigation of the processes causing the patterns in one system (the rocky intertidal community) may shed light on the potential mechanisms involved in the other.

Secondly, experimental studies of zonation patterns may contribute to our understanding of niche widths. Some algae have broad vertical distributions, while others occur only in a narrow band. Factors affecting the widths of the zones are amenable to experimental investigation.

Thirdly, zonation occurs on a biogeographic scale as well as on a local one. A better understanding of the mechanisms affecting local patterns of distribution and abundance may suggest what factors might cause biogeographic species replacement patterns.

This paper reports an experimental investigation of factors affecting zonation of the dominant macroscopic algae in the New England rocky intertidal community. Experiments were designed to evaluate the importance of competition, herbivory, and physical and chemical factors in determining the upper and lower limits of the plants. In particular, the lower limit of the mid zone brown algae *Fucus vesiculosus* L. and *F. distichus* L. ssp. *edentatus* (De la Pyl.) Powell (rockweeds) and the upper and lower limits of the low zone red alga *Chondrus crispus* Stackhouse (Irish moss) were examined. Experiments were designed based on the wealth of the information available on the biology and ecology of *Chondrus* (e.g., reviews in Harvey and McLachlan 1973, Prince and Kingsbury 1973*a*, *b*, Mathieson and Burns 1975).

Study Sites and Methods

Field observations and experiments were carried out at four main sites along the New England shores. Areas inaccessible to most human disturbance were chosen which represented the range of physical and biological conditions existing along these rocky shores. Detailed descriptions of the sites can be found in J. L. Menge (1975) and B. Menge (1976). The four sites (all in USA) fall along a wave exposure gradient. Ranging from least to most exposed, they are Canoe Beach Cove, Nahant, Massachusetts (42°25′N, 70°55′W), Grindstone Neck, Maine (44°10′N, 68°1′W), Chamberlain, Maine (43°56′N, 69°54′W) and Pemaquid Point (same coordinates as Chamberlain). These areas are stretches of continuous rock (usually granite or basalt) with a minimum of cobbles.

Basic information on the distribution and abundance of macroscopic plants and animals was obtained every 2–3 mo at each study area using horizontal transects (described more fully in J. L. Menge 1975 and B. Menge 1976). Estimates of percent cover of organisms were obtained with a ¼-m^2 plexiglass quadrat which had 100 dots plotted from randomly generated coordinates. Estimates of densities of animals were obtained by enumerating organisms present within a ¼-m^2 quadrat.

Space occupiers (plants and sessile animals) exist in three tiers: primary (attached to the substratum), secondary (understory), and tertiary (canopy). The most abundant canopy and understory organisms are three species of fucoid algae and *Chondrus,* respectively. The discoid holdfasts of the fucoids occupy a very small amount of primary space. Their thalli or upright portions pass through understory space and branch out to occupy extensive amounts of canopy space. The holdfast of *Chondrus* forms an often extensive, prostrate red crust on primary space, while the thallus occupies understory space as a shrubby mat. In this paper, "upright *Chondrus*" or just "*Chondrus*" refers only to the erect thallus, while "*Chondrus* crust" refers only to the prostrate holdfast.

The effect of *Chondrus crispus* on the lower limit of *Fucus vesiculosus* or *F. distichus* ssp. *edentatus* was investigated by removing *Chondrus* from flat substrata 1.5–3 m^2 in area in the low zone and monitoring subsequent recolonization patterns. In experiments designated "removal of thallus only," the upright or erect portion of *Chondrus* was scraped off the rocks using putty knives and paint scrapers. This technique leaves the prostrate, encrusting holdfast of *Chondrus* attached to the rock and is designed to mimic natural removals of *Chondrus* thallus that occur during winter storms (Lubchenco and Menge 1978). In the experiments designated "removal of thallus and crust," the thallus was removed by scraping (as above) and then the crust was removed by burning repeatedly with a propane torch and scraping and brushing vigorously. One exception was an experiment at Chamberlain in which a "removal of thallus only" experiment functionally became a "removal of thallus and crust" when the barnacle *Balanus balanoides* settled very abundantly on and completely covered the *Chondrus* crust. *Chondrus*' holdfast is exceedingly tenacious and can be removed only with great difficulty. In replicate experiments done later, sandblasting was used to remove the crust and proved to be a superior technique to burning. Total thallus and crust removals rarely occur in New England, but may occasionally be caused by limpet grazing and scouring by boulders, cobbles, and ice.

The angular transformation has been applied to percent cover data for statistical analyses (Sokal and Rohlf 1969). In no case did transformations alter the outcome of a statistical test. On figures with mean ±95% confidence intervals, percent cover data have been plotted in degrees, then the ordinate labels transformed back into percent cover for easier interpreta-

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TABLE 1. Percent cover of the most abundant space occupiers in New England rocky intertidal zones, summer 1974, at four areas along a wave exposure gradient.*†

		Sites arranged from least to most exposed to wave action (left to right)			
		Canoe Beach Cove	Grindstone Neck	Chamberlain	Pemaquid Point
High Zone					
1°	*Balanus balanoides*	23 ± 8	63 ± 17	93 ± 2	85 ± 7
	Mytilus edulis	13 ± 8	9 ± 12	4 ± 3	5 ± 3
	Crusts‡	43 ± 30	0	0	0
	Bare§	17 ± 11	28 ± 15	3 ± 3	10 ± 7
2°	*Fucus spiralis*	5 ± 7	0	0	0
	Ascophyllum nodosum	66 ± 23	0	0	0
Mid Zone					
1°	*B. balanoides*	6 ± 4	1 ± 2	7 ± 2	8 ± 4
	M. edulis	8 ± 7	26 ± 19	66 ± 9	64 ± 13
	Crusts	37 ± 29	38 ± 26	0	18 ± 22
	Bare	33 ± 25	15 ± 16	3 ± 3	2 ± 3
2°	*Chondrus crispus*	1 ± 1	0	3 ± 2	0
	Ephemeral algae	0	0	12 ± 8	8 ± 9
	Gigartina stellata	0	0	3 ± 3	0
3°	*A. nodosum*	88 ± 6	0	0	0
	F. vesiculosus	5 ± 6	78 ± 21	0	0
	F. distichus	0	0	67 ± 12	3 ± 3
Low Zone					
1°	*B. balanoides*	0	0	0	37 ± 23
	M. edulis	1 ± 2	40 ± 26	52 ± 34	58 ± 21
	Crusts	40 ± 32	11 ± 13	2 ± 2	0
	Bare	3 ± 4	8 ± 12	3 ± 3	0
	C. c. holdfasts	33 ± 13	24 ± 24	43 ± 34	0
2°	*C. crispus*	87 ± 5	68 ± 20	61 ± 28	0
	Ephemeral algae	0	0	0	14 ± 14
3°	*F. distichus*	0	0	1 ± 3	14 ± 14
	Alaria esculenta	0	0	0	44 ± 17

* Mean and 95% confidence interval of 10 quadrats in each zone.

† Any species occupying 10% of the space is included. 1°, 2°, and 3° indicate whether the species occupies primary, secondary (understory), or tertiary (canopy) space.

‡ Crusts include *Verrucaria ericksenii, V. mucosa, Ralfsia verrucosa, R. clavata, Hildenbrandia prototypus, Petrocelis middendorfii, Lithothamnium* spp., *Clathromorphum* sp., *Phymatolithon* sp., and various unidentified red and brown crusts.

§ Bare indicates no visible plants or animals present.

tion. Percent covers of organisms in a single cage and in tables are not transformed.

COMMUNITY STRUCTURE: ZONATION PATTERNS

The type and amount of algae present along New England rocky shores vary along two principal environmental gradients, a vertical tidal (depth) and a wave exposure gradient. The striking zonation of plants along the bathymetric gradient has been described in detail for New England shores (Pearse 1913, Johnson and Skutch 1928, Stephenson and Stephenson 1954, Lamb and Zimmerman 1964, Menge 1976, Lubchenco and Menge 1978, Mathieson, Hehre and Reynolds, *in press,* and Mathieson, Reynolds and Hehre, *in press*), as well as for the eastern coasts of the Northern Atlantic where many of the same species occur (references in Chapman 1957, Lewis 1964, Stephenson and Stephenson 1972). At areas that range from protected to moderately exposed to wave action, four distinct zones along the bathymetric gradient can be easily recognized. The splash zone (Stephenson and Stephenson's "supralittoral fringe," 1972), characterized by blue-green algae, primarily *Calothrix* sp. and the lichen *Verrucaria ericksenii,* lies above the barnacles. The high, mid, and low intertidal zones are composed primarily of the barnacle *Balanus balanoides,* the fucoid algae *Fucus* spp. or *Ascophyllum nodosum* (L.) Le Jolis, and the Irish moss alga *Chondrus crispus,* respectively (Table 1). Each of the zones appears distinct because of the numerical dominance of one type of plant or animal. For the most part, fucoids are present only in the mid zone in New England, as opposed to their presence in both the mid (*F. vesiculosus* and *Ascophyllum*) and low (*F. serratus* L.) zones on European shores. Although there is vertical zonation of the fucoids in New England (e.g., from

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higher to lower mid level at protected areas like Canoe Beach Cove: *F. spiralis* L., *F. vesiculosus*, *Ascophyllum*, and *F. distichus* ssp. *edentatus*), the mid zone is usually dominated numerically by only one of these species (e.g., *Ascophyllum* at Canoe Beach Cove, Table 1). This paper deals primarily with the numerically dominant plant in the mid and low zones.

Table 1 indicates the percent cover of the most abundant space-occupying organisms in each zone at four study areas. Along the horizontal wave exposure gradient, differences within depth zones are apparent when areas are compared. For example, in the mid zone, *Ascophyllum* is the most abundant fucoid present at calm areas (e.g., Canoe Beach Cove). *Ascophyllum* is replaced by *F. vesiculosus* at the next most exposed site (Grindstone Neck) which is in turn replaced by *F. distichus* ssp. *edentatus* at the next most exposed area (Chamberlain). Finally, at the most exposed site (Pemaquid Point), the canopy is nearly absent, with only a 3% cover of *F. distichus*. The most prominent space occupier here is the mussel *Mytilus*. In the low zone *Chondrus* dominates at all but the most exposed areas (Pemaquid Point), where mussels and the kelp *Alaria esculenta* replace it. *Mytilus* is the competitive dominant in both the mid and low zones and outcompetes the algae at exposed sites because its predators are absent or ineffective. At less exposed areas, *Mytilus* and *Balanus* are controlled by their predators in the mid and low zones, thus these levels are characterized by algae (J. L. Menge 1975, B. Menge 1976, 1978*a*, *b*, Lubchenco and Menge 1978). Both the fucoids and Irish moss provide microhabitats for many plants and animals and thus are important structural elements of the community.

This basic pattern of distribution (i.e., zonation along both vertical and horizontal gradients) and abundance (percent cover) of the numerically dominant plants persists throughout the year (Fig. 4 in Menge 1976 for the mid zone; Fig. 1 in Lubchenco and Menge 1978 for the low zone; *Fucus*, *Ascophyllum*, and *Chondrus* are perennials).

Thus the basic pattern observed for macroscopic perennial algae on New England shores is that along the bathymetric gradient, fucoids dominate the mid zone and *Chondrus* the low zone. Along a wave exposure gradient, different fucoids replace one another except at the most exposed sites. Only the bathymetric zonation patterns are considered in this paper. The causes of the horizontal patterns will be considered in a later paper.

Factors Affecting Algal Zonation

Factors affecting the upper limit of a species are considered separately from factors affecting the lower limit. The major portion of this paper focuses on the lower limit of *Fucus vesiculosus* and *F. distichus* ssp. *edentatus*, i.e., the question "Why aren't these two species of *Fucus* usually found in the low zone?" Three factors that could possibly determine the lower limit of these fucoids are differences in (1) immersion time, (2) herbivore effect, and (3) competition intensity. In the following section, each of these is considered and results of experiments testing each are presented. "*Fucus*" refers to both of the two species investigated, unless otherwise noted. "*F. distichus*" refers to *F. distichus* ssp. *edentatus*.

Physical factors: effect of longer immersion time

It has been suggested that many littoral algae, including fucoids, cannot withstand longer immersion times in water and this is the factor setting their lower limits (see discussions and references in Lewis 1964 and Stephenson and Stephenson 1972). A related explanation is that the reduced light intensity at lower littoral levels due to longer immersion times sets lower limits of many algae (Gail 1918). As Connell (1972) points out, there is no direct evidence to support either hypothesis, but neither is there solid evidence to the contrary.

Indirect evidence exists suggesting that the lower limits of fucoids are not determined by death occurring with longer immersion times. Culturing experiments indicate that some fucoids including *F. vesiculosus* grow well under laboratory constant submersion conditions (McLachlan et al. 1971). The physiological tolerance limits of many intertidal algae, including fucoids, are often much broader than the range of conditions under which the species are normally found, indicating these plants should be able to live lower than they do (e.g., Baker 1909, Gail 1918, Klugh and Martin 1927, Chapman 1966, Edwards 1977, Ramus et al. 1977; see also discussion in Connell 1972).

Not only are fucoids normally absent from the low zone in New England, but they are also noticeably absent from tide pools at sites protected from wave action. Experiments indicate that herbivores (which are more abundant and forage longer in pools than out) prevent fucoids from colonizing pools at protected sites (like Canoe Beach Cove). Removal of herbivores and algal competitiors from protected pools results in settlement, growth, and reproduction of *F. vesiculosus* (J. Lubchenco, *personal observation*). At exposed pools where herbivores are rare and ineffective, *F. distichus* L. ssp. *distichus* Powell is often found (Edelstein and McLachlan 1975). Thus, if these species can grow under constant submersion conditions, they should be able to withstand low zone submersion conditions.

Biological factors: effect of herbivory and competition

The biological factors which may affect the fucoids' lower limits, herbivore pressure and competition with other algae, were investigated in experiments designed to separate out the effects of each.

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TABLE 2. Average density (D), size (S), and biomass (B) of herbivores in high, mid, and low zones at four areas along a wave exposure gradient, summer (June and July) 1974.*† Dots indicate categories where no data are appropriate; question marks indicate no data taken.

		Canoe Beach Cove			Grindstone Neck			Chamberlain			Pemaquid Point		
		D	S	B	D	S	B	D	S	B	D	S	B
High	*L. saxatilis*‡	2600	0.6	14.106	7950	0.3	17.928	3450	0.3	7.780	2120	0.3	4.781
	L. obtusata	185	0.8	3.630	3	0.7	0.043	8	0.5	0.054	0	...	...
	L. littorea	1	0.1	0.016	6	0.6	0.031	0	...	...	0	...	...
	Total high	2786	...	17.752	7959	...	18.002	3458	...	7.834	2120	...	4.781
Mid	*L. saxatilis*	18	0.4	0.058	0	...	...	0	...	...	4	0.2	0.006
	L. obtusata	555	0.78	10.277	565	0.63	6.423	14	0.76	0.244	0	...	...
	L. littorea	274	1.49	19.228	59	1.89	8.209	2	0.9	0.033	0	...	...
	Acmaea‡	4	1.1	0.064	15	0.9	0.122	0	...	...	0	...	...
	Lacuna‡	0	...	...	4	0.4	...	0	...	...	0	...	...
	Total mid	851	...	29.627	643	...	14.838	16	...	0.277	4	...	0.006
Low	*L. saxatilis*	0	...	...	0	...	...	0	...	...	0	...	...
	L. obtusata	0	...	...	0	...	...	0	...	...	0	...	...
	L. littorea	649	1.21	25.015	75	1.67	7.308	0	...	...	1	0.7	0.008
	Acmaea	0	...	...	6	1.29	0.165	0	...	...	0	...	...
	Lacuna§	10	0.3	0.047	60	0.58	8.614	1060	0.4	22.223	0	...	...
	Margarites‡§	0	...	...	30	0.32	3.239	0	...	...	0	...	...
	Strongylocentrotus‡	2	?	?	0	...	...ˢ	0	...	...	0	...	...
	Total low	661	...	25.062	171	...	19.326	1060	...	22.223	1	...	0.008
	Total for all zones	4298	...	72.441	8773	...	52.166	4534	...	30.337	2125	...	4.795

* Densities (number per square metre) and size ($\bar{x}$ length in centimetres) are from transect data, 10 quadrats per zone.

† Biomass ($\bar{x}$ grams dry mass per square metre) was estimated from length to dry mass regressions. For *Acmaea, Lacuna, Margarites,* and *Strongylocentrotus,* regressions for these genera from Washington State (Menge 1972 and B. Menge, *personal communication*) were used as approximations.

‡ L. = *Littorina*; all genera listed are coiled snails with the exception of *Acmaea* (limpet) and *Strongylocentrotus* (urchin).

§ *Lacuna vincta* and *Margarites helicina* each have a brief peak density in early summer, then virtually disappear for the rest of the year.

Herbivorous gastropods, in particular the periwinkle *Littorina littorea,* readily consume young *Fucus* plants and under some circumstances prevent *Fucus* from becoming established in the mid zone if the snails are abundant enough (Menge 1975). Herbivory might be expected to be more intense in the low than in the mid zone for three reasons: (1) herbivores may be more abundant in the low zone; (2) different, more effective, or more kinds of herbivores may be present in the low zone; (3) the same kinds, numbers, and sizes of herbivores may be present in both zones, but they feed for longer periods of time in the low zone because the latter is covered by water longer than is the mid zone.

At all of the study sites except Chamberlain, the kinds and biomass of herbivores are roughly equivalent in the mid and low zones (Table 2). Thus if herbivory is more intense in the low zone, (3) seems to be the only reasonable explanation.

Chondrus' domination of low zone space suggested it as a potential competitor with *Fucus*. The following experiments were designed to evaluate the effects of herbivores and competition from *Chondrus* on the lower limits of *Fucus*.

In the low zones of two sites differing markedly with respect to the density of *L. littorea* (Chamberlain, 0/m² and Canoe Beach Cove, 649/m²; Table 2) removals of *Chondrus* and *Chondrus* holdfasts were performed. *Chondrus* occupied 95–100% cover in all treatments prior to removal. The *Chondrus* thalli were scraped off and at Canoe Beach Cove the holdfasts were removed by scraping and burning. In the burning process, all plant crusts (e.g., *Ralfsia* spp., *Hildenbrandia rubra, Petrocelis middendorfii, Verrucaria mucosa,* and encrusting coralline algae) were removed. At Chamberlain, no burning was done, but the same effect of "removing" the *Chondrus* holdfast was fortuitously accomplished when barnacles settled heavily (100% cover) on the cleared patch. These areas (1–3 m²) were cleared during late spring–early summer 1974.

Replicate experiments were initiated at both sites in fall 1974. In spring 1975 additional replicates were performed at Canoe Beach Cove using sandblasting to remove all prostrate portions of *Chondrus* more efficiently.

Results of these experiments are as follows: Where the entire *Chondrus* plant (thallus and crust) is left intact (Fig. 1A and D), no *Fucus* appears. Where upright *Chondrus* thalli, but not encrusting holdfasts, are removed (Fig. 1B and E), new *Chondrus* thalli quickly regrow from the crusts. A few *Fucus* germlings appeared in one of these treatments (Fig. 1E) but quickly disappeared even in the absence of littorines. If the

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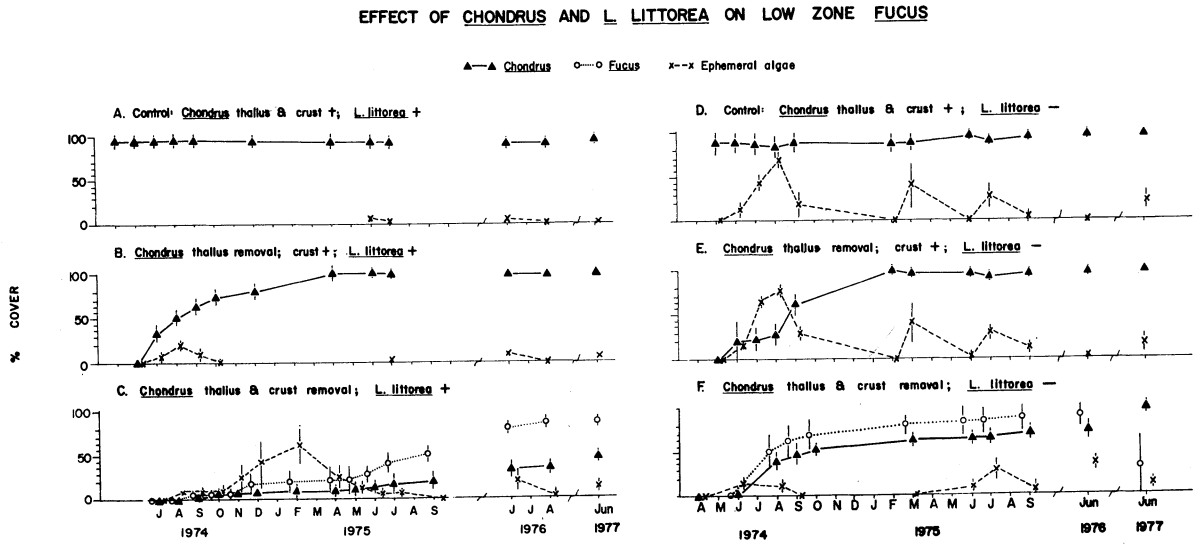


FIG. 1. Effects of competition from *Chondrus* and herbivory by *Littorina littorea* on *Fucus* in the low zone. Each graph represents a separate treatment. A, B, and C, all 3 m², are at Canoe Beach Cove, where *L. littorea* is abundant (Table 2). D, E, and F (2, 2, and 1.5 m²) are at Chamberlain, where *L. littorea* is virtually absent (Table 2). Original percent cover of *Chondrus* (before removals) was 95% in B and F, and 100% in C and E. Points represent means and 95% confidence intervals of angularly transformed percent cover data from 6–12 ¼-m² quadrats at each date. Portions of this figure appeared in Lubchenco and Menge 1978.

Chondrus crust is "removed" (actually or functionally), *Fucus* settles and grows in the low zone. If *L. littorea* is present in such situations (Fig. 1C) *Fucus* is grazed on flat substrata and becomes established only where young germlings are protected by the temporary refuge of crevices. Once these young plants attain a length of 3–5 cm, they are relatively immune from snail grazing (Menge 1975). In the plot indicated in Fig. 1C, enough *Fucus* became established in crevices that subsequent growth resulted in a 90% cover of *F. vesiculosus*. Where *L. littorea* is absent and *Chondrus* is removed (Fig. 1F) *F. distichus* settles very densely and quickly occupies almost 100% cover of space. Thus, *Chondrus* holdfasts, which often form an extensive red crust, may inhibit settlement, attachment, or growth of *Fucus* spp.

Throughout the course of these experiments, various ephemeral algae appear and disappear (Fig. 1). Factors affecting ephemeral algal abundance and their effect on *Chondrus* are discussed in Lubchenco and Menge (1978).

In the low zone experiment at Chamberlain (Fig. 1F), *F. distichus* settled on top of the 100% cover of barnacles that had settled immediately after *Chondrus* blades were removed in April. In early and midsummer, barnacle-eating starfish (*Asterias vulgaris* and *A. forbesi*) invaded this patch and consumed the barnacles, leaving the *Chondrus* holdfast and other encrusting algae. By this time, *F. distichus* holdfasts were firmly attached to all of these crusts. *Chondrus* blades then began regenerating from the holdfasts under the sparse but taller *Fucus* canopy. In July 1974, the mean length (±95% confidence intervals) of *Fucus* in the patch was 0.9 ± 0.1 cm (N = 111). In August, these plants averaged 2.2 ± 1.3 cm long (N = 133), were of normal, healthy color and appearance, and most were reproductive (i.e., they had conceptacles). This 1½-m² patch of almost 100% cover of *Fucus* in the middle of a lush *Chondrus* bed is quite striking (Fig. 2).

The *F. distichus* plants in this experiment persisted throughout 1975 and 1976 (Fig. 1F). By June 1977, however, only a few remained (totaling 19 ± 55% cover; Fig. 1F). Thus, most of the *F. distichus* persisted in the low zone for 2½–3 yr. I surmise that either winter storms removed most of the *Fucus* from the plot in 1977 or that 2½–3 yr is the normal lifespan of *F. distichus*. While the canopy of *F. distichus* was present, *Chondrus* blades slowly perennated underneath, from the crust. By June 1977, *Chondrus* occupied 90 ± 11% cover.

In the experiments at Canoe Beach Cove in which the crust was removed by scraping and burning (Fig. 1C) or sandblasting, *Chondrus* crust slowly colonized the plots, appearing first as small, red discs and spreading over the rocks. The crust originally remaining at the edges of the plot grew inward. Upright *Chondrus* sprouts appeared from all of these crusts and grew into normal *Chondrus* blades. As indicated in Lubchenco and Menge (1978) colonization de novo occurs more slowly than perennation from crusts.

F. vesiculosus in this experiment was still abundant and healthy in June 1977 (Fig. 1E). This persistence of *F. vesiculosus* longer than *F. distichus* might be

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FIG. 2. Photograph of 1.5-m² patch of *Fucus distichus* ssp. *edentatus* in the *Chondrus* zone at Chamberlain, Maine, September 1974, 5 mo after *Chondrus* thallus was cleared and crust was covered by barnacles.

due to a number of factors: (1) *F. vesiculosus* has a longer lifespan than does *F. distichus* (Menge and Lubchenco, *personal observations*); (2) Canoe Beach Cove is more protected from wave action than Chamberlain (Menge 1976) and probably did not receive as much wave surge in winter storms; (3) there may have been differences in the strengths of attachment since *F. vesiculosus* was attached in crevices to what appeared to be bare rock while *F. distichus* was attached to barnacles and *Chondrus* crust.

Regardless of the differences between these two experiments, they both suggest how *Chondrus* retains its dominance in the low zone. Normal disturbances remove only upright *Chondrus,* leaving the *Chondrus* crust, which then quickly perennates (as in Fig. 1B and E; Lubchenco and Menge 1978). *Fucus* does not become established in these cases. If *Chondrus* crust is removed and *Fucus* does become established (as in Fig. 1C and F or any time primary succession is initiated), *Chondrus* grows back underneath the *Fucus* canopy and when those *Fucus* die or are removed *Chondrus* again occupies the space and can prevent *Fucus* from settling. The mechanism of this inhibition warrants further investigation.

Three replicates of the thallus and crust removal experiments begun in fall 1974 and spring 1975 confirm the above results. *F. vesiculosus* and/or *F. distichus* colonize the low zone whenever *Chondrus* uprights and encrusting portions are removed, regardless of whether or not herbivores are present. However, if grazers are absent, *Fucus* colonizes more densely and occupies more space. If grazers are present the amount of space occupied by *Fucus* depends on the abundance of crevices.

Although *L. littorea* is virtually absent from Chamberlain, the area is not totally free of herbivores (Table 2). Isopods, amphipods, and small herbivorous snails like *Lacuna* and *Margarites* are occasionally present in low densities. The effect of their grazing on *Fucus* and *Chondrus* is unknown.

Factors affecting which species of *Fucus* colonized in these experiments include (1) time of year and (2) abundance of each *Fucus* species in the mid zone at a particular area. Time of year translates into which species has zygotes available to settle when *Chondrus* crust is removed. In New England, *Fucus vesiculosus* is reproductive during winter and spring while *Fucus distichus* is reproductive from late spring to early winter. Both species were reproductive when the experiments in Fig. 1 were initiated. Differences in *Fucus* abundance probably result in differences in zygote abundance. *F. vesiculosus* is the more abundant *Fucus* at Canoe Beach Cove while *F. distichus* is more abundant at Chamberlain (Table 1). The more abundant species colonized the low zone at each of these areas (Fig. 1C and F).

Experiments of Burrows and Lodge (1950) indicate that most *Fucus* zygotes are not spread very far from where they were released. All experiments reported here were performed in close proximity to and "downstream" from dense *Fucus* spp. beds, thus numerous zygotes were probably available to settle. *Fucus* settled very densely in all the experiments initiated in late spring–early summer (Fig. 1 experiments and replicates). In replicates initiated in fall 1975 at Canoe Beach Cove, fewer *Fucus* settled. As in the Fig. 1C experiments only those *Fucus* in crevices escaped snail grazing. These plots were not dominated by *Fucus,* but rather contained *Fucus* along with various other algae.

In summary, the experiments reported here clearly indicate that at least *F. vesiculosus* and *F. distichus* are physiologically capable of inhabiting the low intertidal regions. They are normally prevented from

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doing so because they are outcompeted by *Chondrus*. Grazing by *L. littorea* does not prevent recruitment in the low zone by these fucoids, but does seem to slow the rates at which they acquire space. Once *Fucus* becomes established in the low zone, it can persist for a number of years as a canopy above the slowly regrowing *Chondrus*. When the *Fucus* is removed, *Chondrus* crust usually covers a large portion of the substratum and can prevent further recruitment by *Fucus*.

Upper limit of Chondrus

As Connell (1972) indicates, most upper limits of intertidal plants and animals are thought to be set by physical factors. I know of only two examples where biological factors have been implicated. Hoshiai (1964) performed a series of denudation experiments in Japan whose results suggest that competition by one mussel (*Septifer virgatus*) sets the upper limit of another mussel (*Mytilus edulis*). Cubit's (1975) experimental removal of limpets from areas in the high intertidal zone of Oregon suggests that these herbivores set the upper limits of numerous algal species during summer months.

What determines the upper limits of *Chondrus* in New England? Indirect evidence suggests physical factors are of primary importance. As indicated in Table 1, the upper limit of *Chondrus* does not end abruptly at the *Fucus-Chondrus* interface. *Chondrus* occasionally ranges into the mid intertidal in small amounts underneath the fucoid canopy. The fate of these mid-level *Chondrus* plants when the fucoid canopy is removed suggests that desiccation probably sets the upper limit of this alga. In March 1974, two vertical strips of *Ascophyllum* (2 × 12 m and 2 × 18 m) were cleared from the broad mid intertidal zone at Canoe Beach Cove, Nahant. *Chondrus* occupied only 1.7 ± 0.9% (mean ± 95% confidence intervals; $N = 30$ quadrats, each 0.25 m^2) of the understory in these strips, but what was present appeared healthy and normal. Within 2–3 wk after removal of the fucoid canopy, all of the *Chondrus* was completely bleached white, dried out, and dead.

The *Chondrus* in the adjacent control strip (2 × 14 m, no canopy removed) remained healthy and at a constant level of abundance (percent cover at $t = 0$ was 1.0 ± 0.9; at $t = 12$ mo was 1.1 ± 0.9%; $N = 14$ quadrats each time).

The plants that died in the removal strips may have been "shade adapted," and thus may not be expected to survive such a sudden change in the light regime. However, three lines of evidence argue against this interpretation: (1) no new *Chondrus* have appeared in the experimental strips in the 5 yr since the canopy was removed, except where new *Fucus* canopy has grown; (2) *Chondrus* in shallow tide pools in the *Ascophyllum* canopy clearance strips have persisted without their canopy; (3) in the low zone Chamberlain *Chondrus* and crust removal experiments (Fig. 1F), *Chondrus* that had regrown under the dense *F. distichus* canopy did not die or bleach when this canopy was removed in winter 1977.

Chondrus thus appears to be an "obligate understory" species (sensu Dayton 1975) in the mid zone. These experiments suggest that physical factors, probably desiccation, may determine the upper limit of *Chondrus*. Summer observations of bleached individuals of *Chondrus* at the upper fringes of the *Chondrus* bed support this interpretation, although the relationship between bleaching and desiccation is not clear.

Lower limit of Chondrus

The lower limit of *Chondrus* is exceedingly variable and appears to be dependent on several factors. Mathieson and Prince (1973) suggest "The lower limits of distribution for *Chondrus* are determined by water transparency, availability of solid substrate, and competition for space." (see also Mathieson and Burns 1975). However, in some instances, sea urchin grazing appears important. Where sea urchins (*Strongylocentrotus droebachiensis*) are common in the shallow subtidal zone (−0.6 m and deeper), e.g., at Grindstone Neck, July 1974, the *Chondrus* zone ends abruptly at the low intertidal-subtidal interface (−0.6 m) (Table 7 in Lubchenco and Menge 1978). Where sea urchins are rare or absent (e.g., Canoe Beach Cove), the *Chondrus* bed continues down into the subtidal zone to −10 to −20 m (Mathieson and Prince 1973, Table 7 in Lubchenco and Menge 1978).

The sharp lower limit of *Chondrus* at Grindstone Neck was apparently a result of intense sea urchin grazing (Lubchenco and Menge 1978). When urchins disappeared from this area (in winter 1974–1975, for unknown reasons), *Chondrus*' range expanded downward well into the subtidal region. Urchins held in experimental cages below the previous lower limit of *Chondrus* prevented this alga from colonizing (Lubchenco and Menge 1978). Thus, the lower limit of *Chondrus* may sometimes be determined by sea urchins if they are abundant. If these grazers are absent, which was the more usual situation in New England (to at least summer 1977 when my year-round observations ceased), *Chondrus* reaches a deeper lower limit. The factor(s) setting this new limit need to be experimentally investigated.

Discussion

Fucus-Chondrus *competition*

This study suggests that the observed dominance of the mid zone by fucoids and the low zone by *Chondrus* in New England is a result of both physical and biological factors. In the low zone, *Chondrus* is the superior competitor for space and normally excludes fu-

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coids. *Chondrus* thus usually sets the lower limit of *Fucus distichus* and *F. vesiculosus*. The mechanism by which *Chondrus* does so is not understood and needs investigation. Experiments reported above suggest that chemical interference may occur, since even *Chondrus* crust apparently prevents the appearance of *Fucus* germlings. This mechanism appears to be effective primarily against the settlement of *Fucus* zygotes, not against larger, established *Fucus* plants. Once a *Fucus* plant becomes established on a non-*Chondrus* crust surface (bare rock, barnacles, perhaps other algal crusts), it is not obviously affected when it comes into contact with *Chondrus* (crust or uprights). This situation existed in both the *Chondrus* crust and blade removals (Fig. 1C and F and replicates). *Chondrus* crust recruited and/or grew laterally until eventually all *Fucus* were in contact with the crust. An understory of *Chondrus* thus developed beneath a canopy of *Fucus* thalli, with *Chondrus* crusts occupying most of the primary space. Thus "adult-larval" competition (Woodin 1976), or, more properly, adult-zygote competition, appears more important in this interaction than does adult-adult competition. The results of my experiments suggest that once individual *Fucus* become established in the low zone, they usually persist for at least 2½–3 yr. By this time, the *Chondrus* crust has grown to occupy most of the primary space and can prevent further recruitment of new *Fucus*. Therefore even if *Fucus* becomes established in the low zone naturally, *Chondrus* will eventually monopolize most of the space.

Is the low zone an optimal or marginal habitat for *Fucus vesiculosus* or *F. distichus*? Close observation of *F. distichus* settling simultaneously in the mid zone and in the *Chondrus* thallus and crust removal in the low zone at Chamberlain (Fig. 1F) suggests that *F. distichus* grows much faster in the low zone. Laboratory experiments suggest that many *Fucus* species grow faster at simulated lower shore levels than at simulated higher shore levels (Baker 1909). Thus the low zone may represent a "better" physiological habitat for these fucoids, which they are usually prevented from occupying due to competition from *Chondrus*. Dayton (1975) found an apparently similar phenomenon with intertidal kelps zoned along a wave exposure gradient in Washington. *Hedophyllum sessile* is usually outcompeted by *Laminaria dentigera* and *Lessoniopsis littoralis* on more exposed shores which are the apparent optimal habitat of *Hedophyllum*. However, since neither my nor Dayton's study has data or observations on survival or reproductive rates for *Fucus* or *Hedophyllum* in both habitats, definitive statements concerning optimal habitats seem premature.

At protected to moderately exposed sites in New England, *Chondrus* is the superior algal competitor in the low zone. As discussed in Lubchenco and Menge 1978, the domination of this zone by *Chondrus* appears to be primarily a function of two things. First, the encrusting holdfast of *Chondrus* is very tenacious and resistant to removal. Although upright *Chondrus* thalli are often torn loose during storms, especially when matted together by mussels, *Mytilus edulis,* the prostrate crust remains attached to the rock. Herbivores that are capable of grazing the crust (urchins and limpets) are too scarce (Table 2) to have much effect. Second, *Chondrus* readily regrows upright thalli from the crust, thus exhibiting high resilience to disturbance. This combination of characteristics apparently enables *Chondrus* to dominate the low zone at all but very exposed areas.

Biogeographic patterns

What would happen if *Chondrus* did not monopolize the low zone? I have observed two regions outside New England (in Canada and Scotland) where *Chondrus* (or a very similar species, *Gigartina stellata* [Stackh. in With.], Batt., or both) are present but do not dominate the low band. In both regions, *Chondrus* or *Gigartina* crusts appear to be consistently and effectively removed; in both regions, a fucoid co-inhabits the low zone with *Chondrus* or *Gigartina* or both. On the north side of Cape Breton Island, Nova Scotia, ice scouring occurs predictably during the winter (Stephenson and Stephenson 1954). Massive sheets of ice, sufficient to shut down boat traffic during some winters, appear to scrape the shores clean. *Fucus serratus* L. (not present in New England) occupies the low and shallow subtidal zones. On the shores near Millport, Scotland, *Patella* spp. (limpets) are often quite common in the low zone (Connell 1961*a*). Here, *Fucus serratus* co-occurs with *Gigartina* and *Chondrus* in the low regions. Both of these situations might be explained as follows. *Chondrus* and *Gigartina* are prevented from dominating the low zone because their crusts are removed either by ice scouring or intense limpet grazing. Interestingly, the resulting bare space is not taken over by one of the higher fucoids, e.g., *F. vesiculosus* (which is present at both sites) but by a completely different species, *F. serratus*.

Water temperature is the factor commonly cited to "explain" biogeographic distributions of marine algae (e.g., Chapman 1943). Specifically, it is suggested that species like *F. serratus* require cold water to survive and therefore exist only where water temperatures are appropriate. I suggest another alternative for *Fucus serratus*. It occurs from Nova Scotia to Greenland to the European Atlantic shores. This range coincides with either predictable ice scouring or intense limpet grazing, both of which create space in the low zone and perhaps prevent *Chondrus* and *Gigartina* from monopolizing space as occurs in New England. Thus, I suggest disturbance-mediated coexistence may explain this latitudinal zonation pattern. This suggestion needs testing to determine whether the observed cor-

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relation between ice or *Patella* and *Fucus serratus* is causal. I predict that removal of *Patella* from a heterogeneous stretch of shore for a number of years at (e.g.) Millport, would result in *Chondrus* and/or *Gigartina* dominating the area to the exclusion of fucoids.

Zonation patterns in general

In those few systems where zonation patterns have been examined experimentally, the effects of both physical and biological factors have been well demonstrated. In every situation, the "lower limit" (i.e., the edge on the more physically benign side) has been shown to be set by biological factors, either competition or predation. Causes of "upper limits" may not be as clear-cut and thus merit further experimental investigation. There is no a priori reason to believe that systems that have not yet been examined experimentally are any different in these respects. For example, the lower limit of trees along an altitudinal gradient may very well be set by competition or herbivory, and not by the physical factors usually assumed to be the causes.

Width of zones: adaptive compromises

What factors determine the breadth of a species' range? Preliminary experiments on the vertical zonation of British fucoids shed some light on this question. Burrows (1947), Knight and Parke (1950), and Burrows and Lodge (1951) suggest that lower fucoids may outcompete higher ones. They found that all species could live and would initially settle in the lower mid and low zones and that which ones persisted there appeared to be a result of competition. Baker's (1909) fucoid growth rate results coupled with these findings suggest that the lower species simply grow faster than do species normally found higher on the shore. Thus competition between fucoids may be a direct function of growth rates, with faster growing species outcompeting slower growing ones. Recent experiments by Schonbeck and Norton (1978) confirm this interpretation. They demonstrated with transplants and removal of competitors that the lower limit of two fucoids in Scotland is set by competition with other fucoids. Baker's (1909) experiments indicate there is an inverse correlation between growth rate and ability to withstand desiccation. Thus, there appears to be a trade-off between the advantages and disadvantages associated with having a broad distributional range. Species normally found at higher levels grow more slowly (and thus are outcompeted at lower levels) but can withstand desiccation better and thus can grow higher than can the lower species.

These findings emphasize that no single factor operates on a species' range. Instead, the range of each species appears to be a compromise in response to several important selective agents. Thus, a high zone species might be a poor competitor because of its slower growth rate which in turn seems tied to a greater ability to withstand desiccation. Thus, to conclude simply that competition sets the lower limit of that species' range ignores its total biology. Experiments such as removal of *Chondrus* to demonstrate that *Fucus* is physiologically capable of living in the low zone serve the important function of pointing out that *Fucus*' lower limit is not determined by physical factors, but by competition. Thus the selective agent(s) operating on the lower limit can be elucidated. However, focusing on "fundamental" vs. "realized" niches (Hutchinson 1958) can obscure more important questions. As Smith (1975) points out, all natural selection takes place in the realized niche; the fundamental or "no competition" niche (as he prefers to call it) is an imaginary range. Instead of attempting merely to find out how the realized and fundamental niches differ, the real focus should be on what determines the width of the realized niche. We have very little understanding of what sorts of trade-offs exist between various abiotic and biotic factors affecting the upper and lower limits. This is an area of investigation that merits further research. Why, for example, haven't any *Fucus* species evolved a method to cope with *Chondrus*? Is it impossible? Is the selection pressure not great enough? Would a *Fucus* species be spread over too broad a range, i.e., would the trade-offs be stretched too far?

A number of people have suggested that "If *Fucus*' lower limit is set by *Chondrus,* but *Chondrus*' upper limit is determined by desiccation, then isn't it really just physical factors in the end that are important?" If one is interested in understanding the mechanisms involved in the evolution of species' ranges, one must consider the important selective agents acting upon individuals in a population, be they biotic or abiotic. All species are probably affected by both, but in a complex manner we barely understand.

The consideration of species ranges really involves questions about the species' niche widths. The question posed above "What factors determine how broad a zone is?" can be translated into "How broad is a species niche along this axis?" However, considerations of niche widths in current ecological theory and empirical work focus almost totally on competitive interactions to the exclusion of effects of abiotic factors and of consumers (Connell 1975). As the above studies suggest, niche widths are probably a complex function of abiotic factors, competition, predation, herbivory, parasitism, etc. We now have little aid from either theoretical or empirical work in approaching an understanding of factors affecting the width of zones. It seems obvious that a thorough understanding of niche widths will be possible only when we can appreciate what trade-offs are involved.

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